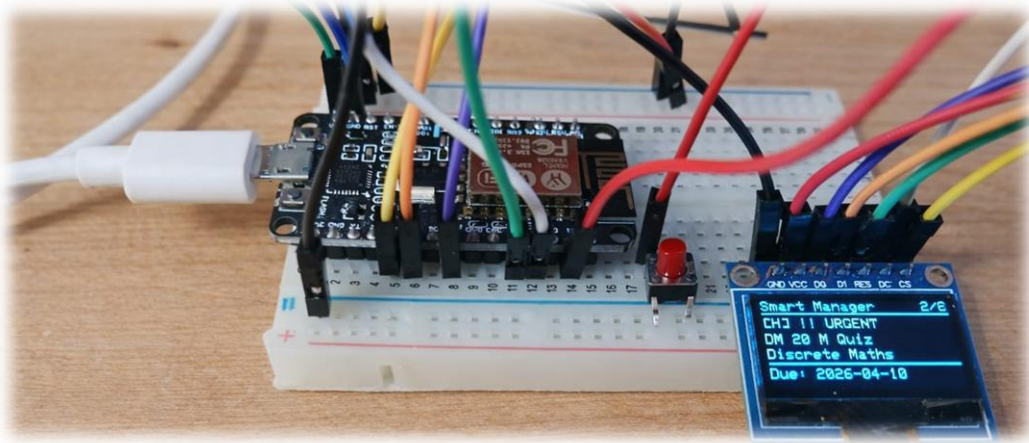


Embedded System Architecture

Proactive Multimodal Crop - Disease Predication with Edge IoT integration S.A.F.E. (Smart Agricultural Forecasting & Edge-computing)



Task:	Ideation : Project Report
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Course	Embedded System Architecture
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Abstract

Traditional precision-agriculture pipelines are reactive: they identify crop disease only after visual symptoms such as dot patches or dark shades have already appeared on the leaf. This project proposes a proactive, multimodal Edge-IoT framework that predicts the probability of a disease outbreak before symptoms are visible, by fusing two complementary signal types: (a) periodic RGB leaf imagery captured at the edge, and (b) a rolling time-series of environmental telemetry — ambient temperature, relative humidity, and soil moisture — which governs the incubation dynamics of fungal and bacterial pathogens. An ESP32-based field node captures RGB leaf images and streams temperature, humidity, and soil-moisture readings over Wi-Fi to a Python backend. A hybrid CNN-LSTM model — a MobileNetV2 branch for spatial image features and an LSTM branch for temporal environmental data — fuses both signal types into a single predictive risk score, surfaced through a Streamlit dashboard. The system is built around low-cost, field-deployable hardware (ESP32 NodeMCU, ESP32-CAM, DHT11, and a capacitive soil-moisture probe) and a lightweight FreeRTOS firmware that streams JSON telemetry over Wi-Fi to a Streamlit-based ingestion API and dashboard, that moves the detection paradigm from “what disease is this” to “is this plant about to get sick.”

1. Introduction

The Internet of Things (IoT) has enabled crop-oriented data sharing and automatic farming solutions, and a representative existing framework — Ensemble Classification and Pattern Recognition for Crop Monitoring System (ECPRC) — demonstrates this well. ECPRC pairs an Arduino Uno with temperature, soil-moisture, and light sensors alongside a camera feed, and applies an ensemble non-linear SVM (ENSVM) and a CNN to classify leaf disease from spectral image patterns, reporting stronger accuracy and recall than K-nearest-neighbours, Naive Bayes, and plain SVM on the same task.

ECPRC, like most systems in this space, is fundamentally reactive — it classifies a disease that has already produced a visible pattern on the leaf. The core objective of this project is to shift that paradigm from detection to prediction, by treating the environmental sensor stream as a time-series that carries an early warning signal, rather than as a side reading only shown on a display. The deliverable is an embedded systems project in the true sense of the term: a field-deployable hardware node (sensing + edge compute), a wireless data pipeline, and a backend multimodal ML system, integrated end-to-end and demonstrated on real or representative data

2. Problem Analysis

2.1 Task 1 limitation — image-only classification

Existing image-only classification pipelines, including ECPRC's ENSVM/CNN approach, identify disease based on visual marks already present on the leaf surface. This is workable once a disease is established, but by design it cannot flag a plant that looks healthy today and is already incubating an infection, because there is no visible pattern yet for the classifier to key on. The missing piece is environmental context: temperature, humidity, and soil condition.

2.2 Task 2 limitation — static classification cannot forecast

Fungal and bacterial infections typically have an incubation period driven by a specific temporal sequence of conditions — for example, sustained high humidity followed by a temperature spike creates favourable conditions for fungal growth well before any lesion appears. A static image classifier has no mechanism to reason about this kind of sequence, because it evaluates a single snapshot rather than a rolling history.

2.3 Proposed solution

A CNN-LSTM architecture that merges spatial data (leaf images, via a CNN) with temporal data (a rolling window of temperature, humidity, and soil-moisture readings, via an LSTM) to forecast outbreak risk before physical damage occurs, rather than only classifying damage that has already happened.

3. Positioning Against the Reference System

Dimension	ECPRC (reference, 2021)	Project SAFE (proposed)
Input	RGB / hyperspectral image only; sensors shown on display, not used by the model	RGB image + temperature + humidity + soil moisture, fused in one model
Task framing	Classifies disease after visible symptoms (ENSVM, CNN, K-NN)	Predicts disease risk before visible symptoms, using a rolling sensor window
Model	ENSVM (subclassifier ensemble) and a from-scratch CNN	MobileNetV2 (pretrained CNN) fused with an LSTM over sensor time-series
Controller	Arduino Uno	ESP32 NodeMCU + ESP32-CAM
Output	Disease notification via GSM alert	Continuous risk score on a live Streamlit dashboard

4. Proposed System Architecture

The system has three stages: an ESP32-based field node that senses and transmits, a Python backend that fuses the two data modalities, and a dashboard that presents the resulting risk score.

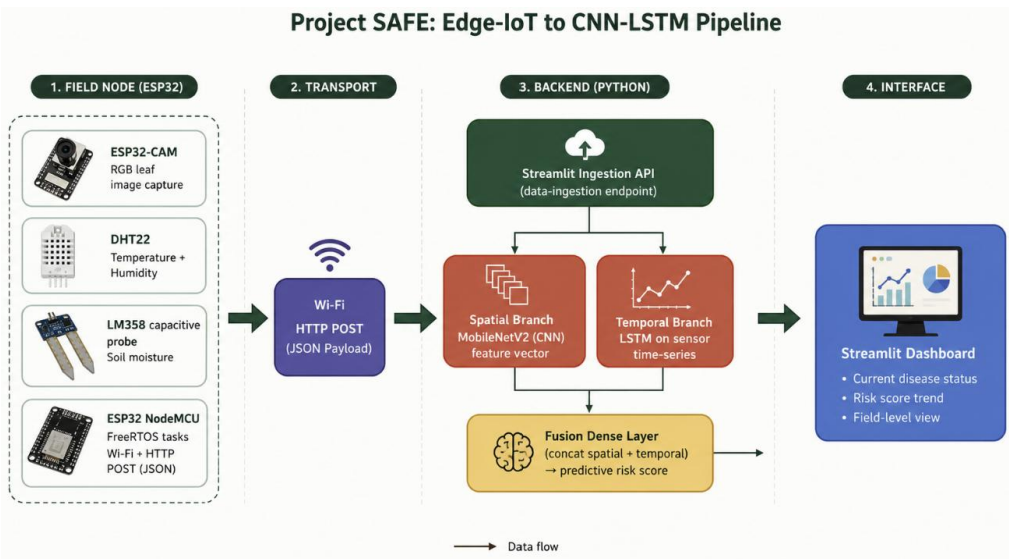


Fig. 1 — Field node (ESP32 + sensors) → Wi-Fi/HTTP → backend fusion model → Streamlit dashboard.

4.1 Field node

The ESP32-CAM captures RGB leaf images on a fixed interval; the DHT11 reports ambient temperature and humidity; the LM358-based capacitive probe reports soil moisture. All three feed into the ESP32 NodeMCU, which runs lightweight FreeRTOS tasks to poll the sensors and transmit a JSON payload to the backend over Wi-Fi via HTTP POST.

4.2 Backend fusion model

A Streamlit-hosted ingestion endpoint receives the incoming payloads. A pretrained MobileNetV2 (with its base layers frozen) extracts a spatial feature vector from each leaf image; an LSTM network processes the corresponding rolling window of sensor telemetry to capture the temporal pattern. Both outputs are concatenated into a dense fusion layer that computes a single predictive risk score.

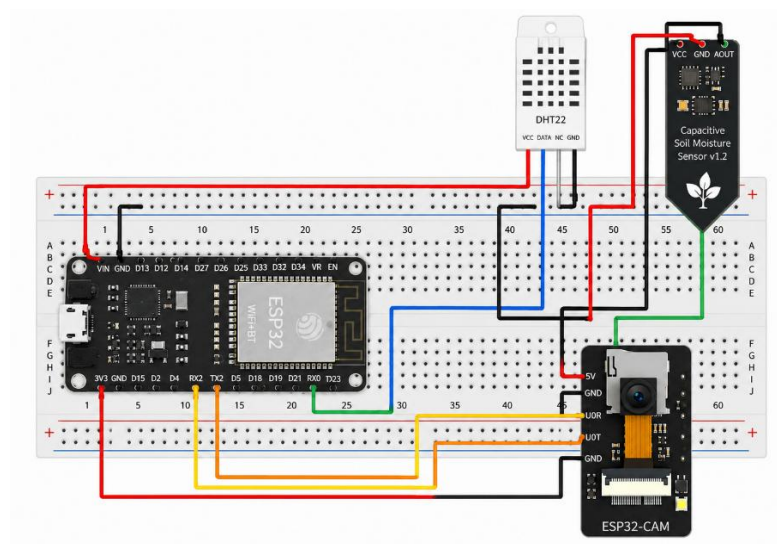
4.3 Dashboard

The same Streamlit application serves the interactive frontend, showing the current risk score and its trend over time, so the output is a continuously updated forecast rather than a one-off classification.

5. Hardware Requirements

The hardware requirement of materials is intentionally minimal, favouring well-supported modular components over custom PCB fabrication to keep the build achievable within a semester timeline.

Component	Role	Notes
ESP32 NodeMCU	Main microcontroller	Native Wi-Fi + HTTP client; runs the FreeRTOS sensor-polling tasks
ESP32-CAM	Imaging module	Captures high-resolution RGB field images of the leaf
DHT22 module	Atmospheric sensor	Ambient temperature and humidity
LM358-based capacitive probe	Ground sensor	Soil moisture reading
Breadboard + jumper cables	Prototyping	Standard prototyping hardware, Male-to-male and male-to-female jumpers
Regulated 5V power supply	Power	Powers the ESP32 and sensor set



6. Software & Machine Learning Architecture

Layer	Tool / Framework	Purpose
Edge firmware	C++ (FreeRTOS on ESP32)	Lightweight asynchronous tasks to poll sensors and transmit JSON over HTTP POST
Spatial branch	Python / TensorFlow, MobileNetV2 (pretrained, base layers frozen)	Extracts spatial feature vectors from RGB leaf images
Temporal branch	Python / TensorFlow, LSTM	Processes the sequenced temperature/humidity/soil-moisture telemetry
Data fusion	Dense neural layer	Concatenates spatial + temporal outputs into a final predictive risk score
Interface & API	Streamlit	Single application handling both the data-ingestion endpoint and the interactive dashboard

7. Implementation Plan

1. (Hardware & Firmware): Wire the DHT22 and LM358 sensors to the ESP32 on the breadboard. Write the C++ firmware to poll sensors, implement the Smart Trigger logic, and transmit data packets over Wi-Fi.
2. (Data Strategy): Aggregate RGB leaf images from open-source dataset. Develop Python scripts to generate synthetic, realistic time-series weather and soil data corresponding to healthy vs. diseased crop states.
3. (Model Training): Construct the CNN-LSTM pipeline. Train the LSTM and fusion layers using the synthetic temporal data. Wrap the finalized machine learning model in a Python FastAPI backend.
4. (Integration & UI): Deploy the Python backend and Streamlit dashboard; establish the live connection between the ESP32 hardware and the predictive model to validate end-to-end functionality.

8. Evaluation Plan

- Classification quality: precision, recall, and F1 of the fused model on held-out PlantVillage images, benchmarked against a MobileNetV2-only (no environmental input) baseline to quantify what the temporal branch adds.

- Forecasting signal: on the synthetic temporal data, check whether the risk score rises measurably in the days before the labelled “onset” point, rather than only at the onset point itself — this is the evidence that the system is predicting rather than only classifying.
- End-to-end latency: time from sensor reading on the ESP32 to an updated risk score on the dashboard, to confirm the pipeline is responsive enough for a live demo.
- Smart Trigger accuracy: false-positive rate (camera wakes when risk was actually low) and false-negative rate (camera stays dormant during a genuinely high-risk window) of the threshold logic, since this directly trades off power/bandwidth savings against missed detections.

9. Future Scope

The multimodal transformers, CNN-LSTM hybrids, and graph neural networks are some possible candidate models, and LSTM, GRU, Temporal Fusion Transformer, and reinforcement learning. This project deliberately scopes its Phase-1 build to the CNN-LSTM hybrid, since it is the most tractable option to implement, train, on modest hardware and within available data. The remaining candidates are natural next steps rather than discarded ideas:

- Multimodal transformer fusion (in place of the concatenation-based fusion layer) once enough multimodal field data exists to justify the extra model complexity.
- Temporal Fusion Transformer in place of the LSTM branch, for interpretable multi-horizon forecasts and native handling of weather-forecast inputs alongside observed sensor history.
- A graph neural network across multiple field nodes, if the deployment grows beyond a single sensor station, to model how disease risk propagates spatially across a field.
- A reinforcement-learning layer on top of the risk score, to move from “here is the risk” to a concrete recommended action (e.g., irrigate / spray / no action).

10. Conclusion

Project S.A.F.E (Smart Agricultural Forecasting & Edgecomputing) keeps the sensing and IoT foundation that make ECPRC-style systems practical to build on a student budget and timeline, while directly addressing its main limitation: detection only after a disease is already visible. By fusing an ESP32-CAM image stream with a rolling window of temperature, humidity, and soil-moisture readings through a CNN-LSTM model, the project aims to demonstrate — even at a proof-of-concept scale — that environmental time-series data carries an early-warning signal that image data alone does not.